

Building reliable data pipelines and clear system observability with Python, Docker, Kubernetes, CI/CD, Apache Kafka, Grafana, and Prometheus — currently expanding into Agentic AI engineering.

Open to Work

Full-Time

Remote

Part-Time

Project-Based

Contract

About

Hello, I'm **Arash Zare**, a **Data Engineer** specializing in system observability and container-based infrastructure. I hold a Master's degree in Computer Engineering from Shiraz University, and my project approach combines data engineering principles with DevOps stability standards to build reliable, scalable pipelines.

Currently, alongside my operational work, I am deeply engaged in studying and developing skills in **AI Engineering**. I am working on personal projects focused on **self-hosted**



Data Engineering

Architecting scalable containerized infrastructure (K8s/Docker) to power robust Data Engineering pipelines and Agentic AI ecosystems.



Big Data Systems

Proficient in Hadoop, Kafka, ClickHouse for large-scale data processing



Cloud Infrastructure

Experience with cloud systems and infrastructure management at scale



System Monitoring

Skilled in Grafana, Prometheus, and ELK Stack for system observability

Skills & Expertise

Programming & Development



Python
Data processing, automation, API integration, and ETL scripting



Go (Golang)
Building efficient microservices and tools



Bash Scripting
Linux automation, deployment scripts, cron jobs, and log processing



Networking
TCP/IP, DNS, firewalls, routing fundamentals and service connectivity



SQL
SQL queries, joins, aggregations, CTEs, and relational data modeling

Big Data & Analytics



Apache Kafka
Real-time data streaming, consumer groups, topic design, and monitoring



ClickHouse
Analytics data modeling, ingestion, and query optimization



Apache Hadoop
Working with HDFS and Parquet files in distributed data environments



Data Pipeline Design
Designing reliable ETL/ELT workflows, scheduling, validation, and monitoring



Apache Spark
Large-scale data processing frameworks

DevOps & Infrastructure



Docker
Containerization, Docker Compose, image optimization, and service deployment



GitLab CI/CD
Designing CI/CD pipelines for testing, building, and automated deployment



Kubernetes
Container orchestration and cluster management



Ansible
Infrastructure provisioning and configuration using playbooks and roles



Terraform
Infrastructure provisioning with providers, modules, and state management

Monitoring & Observability



Grafana
Building operational dashboards, alerts, and data visualizations



Prometheus
Metrics collection, PromQL, alerting rules, exporters, and monitoring design



System Monitoring
Linux performance, service health, capacity tracking, and incident troubleshooting



ELK Stack
Log collection, indexing, search, and visualization with Elasticsearch and Kibana



Ceph & MinIO
S3-compatible object storage, bucket management, and data platform integration

Work Experience

Data Engineer

Derak Cloud Company

2022 – Present · 2 yrs 6 mos

Shiraz, Fars, Iran

Developed and maintained data infrastructure and cloud systems. Specialized in building scalable data pipelines and implementing DevOps practices for improved system reliability and performance.

- Led the Cluster Health Check project, improving system performance and stability
- Implemented containerization using Docker for efficient deployment and scalability
- Built robust data processing pipelines using big data technologies
- Managed cloud infrastructure and monitoring systems

Graduate Teaching Assistant — Cloud Computing

Shiraz University

Fall 2023 · 1 semester

Shiraz, Fars, Iran

Teaching assistant for the advanced Cloud Computing course under Prof. Dr. Khunjush, Department of Computer Engineering.

- Mentored students through cloud computing concepts, virtualization, and distributed systems
- Designed hands-on labs on containerization (Docker), virtualization, and cloud architectures
- Ran problem-solving sessions and technical Q&A for distributed systems assignments

Freelance Software & Linux Systems Developer

Self-Employed / Academic Projects

2020 – 2022 · 2 yrs

Shiraz, Fars, Iran

Delivered web applications and system automation for academic and small business clients. Focused on Python/Django, Linux administration, and automation scripting.

- Built modular web apps with Django, Flask, and PostgreSQL/SQLite
- Configured and maintained Linux servers (Ubuntu/CentOS): SSH, Nginx reverse proxy, services
- Automated recurring workflows and log/data collection with Bash and Python
- Containerized projects with Docker for reproducible deployments

Featured Projects

Multi-Cluster Big Data CLI Automation Suite

A modular CLI automation framework developed at Derak Cloud using Bashly and Docker Compose. It streamlines lifecycle management, configuration synchronization, and cross-environment orchestration for ClickHouse, ELK Stack, Hadoop (HDFS/YARN), and PrestoDB clusters.

Bashly Docker Compose kafka ClickHouse Hadoop ELK

Health Check Cluster Monitoring System

Comprehensive cluster health monitoring system built at Derak Cloud. Improved system performance and stability through real-time monitoring and alerting.

Grafana Prometheus Kubernetes Docker Python

NetFlow & Log Forwarding Stack

Scalable log forwarding and network traffic monitoring pipeline: syslog-ng for log collection, goflow2 for NetFlow/sFlow/IPFIX, streamed into RabbitMQ.

syslog-ng RabbitMQ goflow2 NetFlow Docker Linux

Project to Reduce Military Service Duration – Deep Learning with PyTorch for the Police Force

Deep learning models using PyTorch for data analysis and pattern recognition supporting law-enforcement operations.

PyTorch Python Deep Learning Computer Vision

In-Progress AI Projects

Self-Hosted Enterprise RAG & AI Chatbot

A fully containerized, privacy-focused local RAG pipeline and conversational system. Deployed using Docker, Ollama (running local LLMs and Nomic embeddings), and Qdrant vector database with sliding-window chunking to eliminate external API dependency and cost.

Ollama Qdrant Docker Python Local LLM Vector DB

[View on GitHub](#)

Agentic AI & Autonomous Workflows

Designing autonomous multi-agent systems and agentic execution loops. Implementing Model Context Protocol (MCP), tool-augmented reasoning, and dynamic task orchestration leveraging LangChain and LangGraph patterns.

LangChain LangGraph MCP Agentic AI Python

[View on GitHub](#)

Publications



A two-tier multi-objective service placement in container-based fog-cloud computing platforms

Cluster Computing (Springer) · Nov 2023

Research on optimizing service placement in container-based fog-cloud computing environments using multi-objective optimization techniques.